



FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECJ3744 ENTERPRISE SYSTEMS DESIGN & MODELLING

SECTION 02

Technical Report

GROUP PROJECT (10%)

Using AI in ERP for Demand Forecasting

LECTURER: Dr Noorminshah binti A.Iahad

STUDENT NAME	MATRIC NO
CHAU YING JIA	A23CS0213
GUI KAH SIN	A23CS0080

Executive Summary

This technical report explores how Artificial Intelligence (AI) can be incorporated into Enterprise Resource Planning (ERP) systems for demand forecasting. Traditional ERP forecasting methods mainly depend on historical data, which often fail to adapt to rapid market changes and unpredictable consumer behaviors. By integrating AI and Machine Learning (ML), ERP systems have transformed into intelligent resources capable of analyzing both internal business data and external market signals to support better strategic planning and decision-making with more accurate forecast.

The report reviews key literature, outlines how AI forecasting functions within ERP environments, and emphasizes its practical implementations in sectors such as e-commerce, retail, and manufacturing. It identifies key advantages, including improved accuracy, reduced inventory waste, and data-informed decision-making, while facing challenges like data governance, integration with legacy ERP systems, and user adaptation. The discussion also included the future trends of AI-driven ERP systems, such as Explainable AI (XAI), which increases transparency, confidence, and comprehension in AI-generated predictions within ERP environments. In conclusion, the report recommends gradual deployment, robust data management, and well-structured employee training to fully recognize AI's capability in modern ERP-based forecasting.

Introduction

Enterprise Resource Planning (ERP) systems are the main core of modern organizations. It integrates key business processes across finance and human resources to supply chain management. In this report, we will be focusing on demand forecasting, which provides strategic suggestions regarding product inventory, production planning, and resource allocation. In fact, forecasting within an ERP systems relied on conventional statistical models that only look at the past sales record as a reference. However, they are hard to harmonize with volatile marketplace scenarios, shifting consumer behaviour and unforeseen externalities like supply chain breakdowns (Choudhuri, S. S., 2024). Using historical data only can result in committing huge mistakes, leading to overstocking at excess cost or missed sales because of stockouts, impacting directly the bottom line.

Machine Learning (ML) and Artificial Intelligence (AI) have brought about a huge change in how business organizations manage demand forecasting and inventory so that they can make accurate predictions and improve decision-making. With the introduction of AI features into ERP solutions, business companies can escape from the drawbacks of traditional forecasting. Artificial intelligence capabilities are superior to understand massive datasets in real-time, not only historical sales but unstructured data because of real-time variations, adapt to market volatility and predict future trends, facilitating better future planning (Aggarwal, P. et al., 2023). This enhanced capability allows managers to make quicker and better strategic decisions based on data-driven insights.

The aim of this technical paper is to bring about a wide-ranging discussion on incorporating AI in ERP systems with the intention to utilize especially for forecasting demand. The goal is to establish how AI technology enhances levels of forecasting and enables decisions to be more dynamic and well-informed. This paper will focus on the explanation of the mechanisms of AI, material benefits, implementation issues, as well as recommendation on

the existing use. We will be explaining a bare-fact account of how this convergence of technology is changing business processes and conferring a strategic advantage on those businesses and firms that managed to use it effectively.

Background / Literature Review

Over the past years, the infusion of AI into ERP systems has led to a new breed of AI-powered ERP solutions. Rather than substituting prior ERP features, AI improves analytical capabilities within a system, especially relevant to demand forecasting, as shown by learning from vast, multidimensional datasets (Nguyen, 2023). Traditional ERP module forecasting tools tend to utilize static, rule-based algorithms without addressing the volatility of modern supply chains. According to the current research, learning models like machine learning (ML) and deep learning (DL) overcome these shortcomings by generating forecasts that are both adaptive and probabilistic (Kaur & Singh, 2024).

In order to increase accuracy, hybrid forecasting frameworks that incorporate AI and traditional statistical models have been proposed in recent studies. For instance, Ashok (2024) demonstrated that deep learning models like Long Short-Term Memory (LSTM) networks outperform ARIMA models when trained on ERP data enriched with external variables like promotions, holidays, and regional economic indicators. On that note, findings by Mohammed and Mandal (2024) reinforce the potential of AI-driven forecasting models instils greater value in decision-making reliability within a suitable enterprise system.

Given the injection of AI-enabled forecasting modules from several ERP vendors, including SAP and Oracle, systems can facilitate real-time analytics and decision support. (Gartner, 2024). Self-retraining becomes a system function's foresight when a new set of data is received, leading to a continuous learning process. However, certain scholars warn that the benefits of AI are conditional upon robust data governance and adoption within its employees. Poor data quality or lack of interpretability can also contribute to decision-making failures and reduce forecasting effectiveness.

Despite increasing adoption, the literature identifies notable research gaps. Discussion on concerns, including the challenges of explanation, scalability, and sustainability within AI-enabled ERP forecasting, is still a rarely shared concern. Furthermore, limited empirical work has been conducted in Malaysia, where data maturity and infrastructure vary compared to global enterprises of high calibre. Consequently, future studies are encouraged to explore localised implementation frameworks and strategies for the mature integration of intelligent ERP technologies tailored to regional contexts.

Analysis and Discussion

The combination of Artificial Intelligence (AI) in Enterprise Resource Planning (ERP) systems completely transforms the dynamics of demand forecasting at its root level. This innovation goes beyond simple historical analysis to a dynamic, data-driven predictive process. The inner strength of this union is that AI could process huge, diverse datasets and learn progressively, thereby providing a greater and more accurate view of what future demand will be.

How AI-Powered Forecasting Works in an ERP Environment

The functioning of an AI-based forecasting system within an ERP environment is a recursive process (Figure 1). It begins with data collection, where the system compiles structured internal data (e.g. historical sales, stock) and combines it with enormous amounts of unstructured external data (e.g. social media, competitor prices). Next, sophisticated Machine Learning (ML) models will work on this combined dataset to reveal all hidden patterns and non-linear relationships between datas which are invisible in traditional statistical tools. Instead of a point prediction, the system provides a probabilistic prediction, offering a set of potential demand scenarios and their probabilities (Ch, Rajitha., 2025). It facilitates more strategic, risk-based planning by the managers. Last but not least, the last step is a continuous feedback loop. As new sales data coming in, the AI compares this against its predictions and re-trains its models automatically, learning from any discrepancies and getting progressively more accurately.

Case Examples and Applications

Many industries in Malaysia have demonstrated the incorporation of Artificial Intelligence (AI) into Enterprise Resource Planning (ERP) systems. For example, two of the most competitive e-commerce sites, Shopee and Lazada use artificial intelligence (AI) to examine past sales data and buyers' browsing habits to predict demand for particular products during significant events like Hari Raya or 11.11 (Li et al., 2025). Major hypermarkets in the Fast-Moving Consumer Goods (FMCG) industry, like Aeon, can use AI to optimise stock levels for perishable goods while minimising waste by comparing point-of-sale data with local weather and holidays. Other than that, national automakers like Proton or Perodua use AI to predict demand for various car models while accounting for pros and cons related to the economy.

Advantages and Disadvantages

Integrating AI technology into ERP systems significantly enhances the quality of forecasting by identifying hidden, non-linear patterns within massive datasets. Machine learning algorithms aims to deliver the real-time data analysis, which enables organizations to respond quickly to market fluctuations and minimize inventory costs (Kaur & Singh, 2024). AI-driven forecasts within the ERP systems ensure the accuracy which help to prevent overstocking, optimize resource allocation across integrated business systems, and provides enterprises with more strategic and well-planned capabilities. Furthermore, AI enhances employee productivity and efficiency by reducing reliance on manual data analysis. Automated forecasting eliminates human bias and the occurrence of errors to avoid any subjective decision-making. However, AI implementation in ERP necessities high-quality and consistent data because poor data governance can lead to inadequate system outcomes. Moreover, insufficient employee training may cause misunderstanding of results too. Thus, these pros and cons underscore the need for careful and well-planned in AI adoption.

Challenges and Future Trends

Applying AI in ERP forecasting brings several challenges. Integration complexity remains to be one of the main barriers, as majority of organizations still rely on legacy ERP infrastructures that are incompatible with modern AI frameworks. Besides, the integration possesses the process is extremely costly and time-consuming as it required specialized expertise and continuous technical maintenance. Various challenges emerge with the governance of data This is pertinent in the sense that AI models may struggle to generate accurate and consistent results

if the data is standardized and validated. Additionally, resistance to change and limited technical understanding may inhibit adoption, especially when employees distrust the results of algorithm or misinterpret the forecasts.

Looking ahead, future trends, for instance, Explainable AI (XAI), are gaining prominence in AI forecasting demand. With the increasingly complexity of AI models, understanding the logic behind AI decisions taking place in an ERP system is becoming vital. According to GeeksforGeeks (2023), XAI enhances transparency by clarifying how models make these predictions, and therefore making the results more interpretable for users, while Ruiqi et al. (2025) emphasizes that explainability improves trust and accountability, thus enabling planners to trace influential variables in advance as well as validate the forecasting produced in the ERP systems effectively.

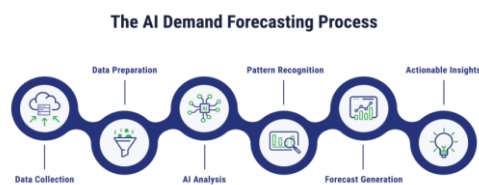


Figure 1: The AI Demand Forecasting Process

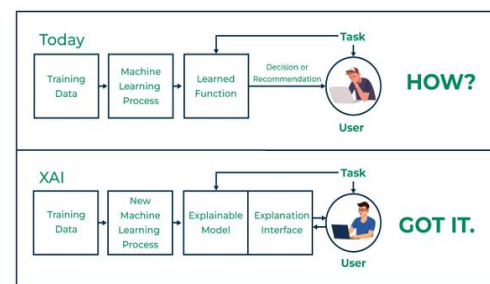


Figure 2: Explainable Artificial Intelligence Concept (GeeksforGeeks, 2023)

Recommendations

For organizations to be able to effectively apply AI in ERP towards demand forecasting, two main methodologies should be implemented. These methodologies should approach both the deployment of the system operationally as well as the business requirements required.

First of all, an iterative deployment approach should be used rather than implementing the new AI system in the entire organisation at the same time. There must be a Proof-of-Concept (PoC) model on a small data domain initially to reduce risks. The basic starting point is to develop strong ETL/ELT pipelines (Figure 2) to process, clean, and transform history ERP data and external data streams to have the better quality of the training dataset. This PoC prototype attempts to train and develop a basic forecasting model, whose performance is extensively tested with metrics like Mean Absolute Error or Root Mean Square Error (Wen et al., 2024). A successful PoC delivers the empirical proof required to justify extending data architecture to enterprise-wide production deployment.

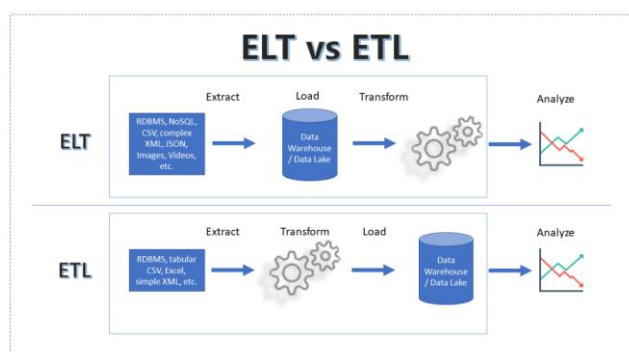


Figure 3: difference between ELT and ETL pipelines

Secondly, effective implementation of AI in ERP also depends on investment in high-quality data as well as upskilling among employees. AI models are completely dependent on what data they learn on. Erroneous or missing data points in the data will lead to incorrect AI forecast output. Therefore, a critical step is to establish the data governance, which involves implementing rules such that all the data must be clean, accurate, and trustworthy (Hausenloy et al., 2024). Besides, organizations need to train staff members by training programs in a bid to attain their business goals. Managers who have relied on manual forecasting would lack of confidence in the new AI analysis. Acquiring data literacy skills enables them to understand how AI develops its insights. The training program builds confidence among employees and AI systems which in turn empowers employees to utilize AI in making improved strategic decisions.

Conclusion

A revolutionary step toward improving the precision of demand forecasting and operational decision-making is the incorporation of Artificial Intelligence (AI) into Enterprise Resource Planning (ERP) systems. By moving beyond static statistical models, AI enables organizations to analyze vast, complex datasets in real time, optimize workflow, discover hidden patterns, and respond swiftly to market fluctuations. This synergy decreases costs, enhances productivity, and reinforces competitiveness.

Nevertheless, achieving these benefits necessitates overcoming persistent challenges such as improper data governance, integration with legacy systems, and user resistance. Implementation of success relies on the establishment of standardized data frameworks, continuous model validation, and comprehensive employee training. In addition, future trends such as Explainable Artificial Intelligence (XAI) will be crucial to ensure transparency, interpretability, understanding, and trust in AI-generated forecasts.

Ultimately, AI-enabled ERP systems represent a major shift toward more intelligent, data-driven, and efficient enterprise management. Businesses will have a long-term edge in the business environment that is becoming more and more dynamic if they make strategic investments in reliable, high-quality data, moral AI procedures, and human-machine collaboration.

References

1. Choudhuri, S. S. (2024). AI-Driven Supply Chain Optimization: Enhancing Inventory Management, Demand Forecasting, and Logistics within ERP Systems. *International Journal of Science and Research (IJSR)*, 13(3), 927-933.
2. Aggarwal, P., & Aggarwal, A. (2023). AI-Driven Supply Chain Optimization in ERP Systems Enhancing Demand Forecasting and inventory Management. *International Journal of Management, IT & engineering*.
3. Ch, Rajitha. (2025). From Data to Decisions: AI-Powered Forecasting in ERP Environments. https://www.researchgate.net/publication/394925139_From_Data_to_Decisions_AI-Powered_Forecasting_in_ERP_Environments
4. Li, Y., Zheng, R., Hao, Y., He, J., & Zhang, X. (2025). The Impact of Artificial intelligence on ecommerce Consumer Behavior: A Comparative analysis. In *Advances in economics, business and management research/Advances in Economics, Business and Management Research* (pp. 60–71). https://doi.org/10.2991/978-94-6463-835-6_8
5. Wen, Xinyu & Liao, Jiacheng & Niu, Qingyi & Shen, Nachuan & Bao, Yingxu. (2024). Deep learning-driven hybrid model for short-term load forecasting and smart grid information management. *Scientific Reports*. 14. 10.1038/s41598-024-63262-x.
6. Hausenloy, J., McClements, D., & Thakur, M. (2024, December 5). *Towards data governance of frontier AI models*. arXiv.org. <https://arxiv.org/abs/2412.03824>
7. Nguyen, T. T. H. (2023). *Applications of Artificial Intelligence for Demand Forecasting*. *Operations and Supply Chain Management Journal*, 16(4), 380–392. <https://doi.org/10.31387/oscm0550401>
8. Kaur, H., & Singh, J. (2024). *Machine Learning Approaches for Forecasting in ERP Systems: A Review*. *Springer Nature Computer Science*, 5(3), 312–326. <https://doi.org/10.1007/s42979-024-02518-2>
9. Ashok, P. P. K. (2024). *AI-Powered Demand Forecasting in ERP: A Comparative Study of ML Algorithms*. *International Journal of Computing and Engineering*, 5(2), 1–11. <https://doi.org/10.47941/ijce.3106>
10. Mohammed, I., & Mandal, J. (2024). *Forecasting Accuracy through Machine Learning in Supply Chain Management*. *International Journal of Supply Chain Management*, 9(6), 8–24. <https://doi.org/10.47604/ijscm.3074>
11. Gartner. (2024). *How AI is Transforming Supply Chain Management*. Retrieved from <https://www.gartner.com/en/supply-chain/topics/supply-chain-ai>
12. GeeksforGeeks. (2023). *Explainable Artificial Intelligence(XAI)*. GeeksforGeeks. <https://www.geeksforgeeks.org/artificial-intelligence/explainable-artificial-intelligencexai/>
13. Hayes, M., & Downie, A. (2024, July 8). *Artificial Intelligence in ERP | IBM*. Ibm.com. <https://www.ibm.com/think/topics/ai-in-erp>
14. Morrison, C. (2023, November 8). *AI in ERP: The Next Wave of Intelligent ERP Systems*. Top10erp.org. <https://www.top10erp.org/blog/ai-in-erp>
15. Z Ruiqi, C. Christensen, B. Zarrin, P. Bækgaard & T. S. Alstrøm. (2025). *Towards Trustworthy AI in Demand Planning: Defining Explainability for Supply Chain Management*. In Proceedings of the SciTePress Conference. Available at: <https://www.scitepress.org/Papers/2025/133159/133159.pdf>

Appendix

Group 12

BacklogTeam capacityCurrent iterationRoadmapMy itemsNew view

Filter by keyword or by field

Todo8 / 5Estimate: 0

This item hasn't been started

Assignment-1-SECP3744-Sec02 #53

Executive Summary

Assignment-1-SECP3744-Sec02 #55

Background / Literature Review

Assignment-1-SECP3744-Sec02 #57

Analysis and Discussion

Assignment-1-SECP3744-Sec02 #58

Recommendations

Assignment-1-SECP3744-Sec02 #59

Conclusion

Assignment-1-SECP3744-Sec02 #60

References

+ Add item

In Progress2 / 5Estimate: 0

This is actively being worked on

Assignment-1-SECP3744-Sec02 #63

Discuss on our topic and do some research

Assignment-1-SECP3744-Sec02 #54

Introduction

+ Add item

Done1Estimate: 0

This has been completed

Assignment-1-SECP3744-Sec02 #56

Task Distribution

+ Add item

Group 12

BacklogTeam capacityCurrent iterationRoadmapMy itemsNew view

Filter by keyword or by field

October 2025

1234567891011121314151617181920212223242526272829303112

1Task Distribution #56

2Introduction #54

3Discuss on our topic and do some research #63

4Recommendations #58

5Background / Literature Review #55

6Analysis and Discussion #57

7Conclusion #59

8Executive Summary #53

9References #60

10Appendix / Logbook #61

11Final Check & Submission #64

+ Add item

Gemini

2.5 Flash

my topic chosen is Using AI in ERP for Demand Forecasting, could you help me out on this? i am still a little blur on this

That's an excellent topic! "Using AI in ERP for Demand Forecasting" is highly relevant to Enterprise Systems Design & Modelling.

To help you get started on your technical report, let's break down the core components of your chosen topic and how you can structure your paper around it.

1. Understanding the Topic: AI in ERP for Demand Forecasting

Your topic focuses on how advanced Artificial Intelligence (AI) technology is integrated with Enterprise Resource Planning (ERP) systems to perform **demand forecasting**.

ERP (Enterprise Resource Planning): This is the system that manages and integrates a company's core business processes (finance, human resources, supply chain, manufacturing, etc.). It is the source of all the historical sales data, inventory levels, and other internal factors.

AI (Artificial Intelligence): This refers to the systems (like Machine Learning algorithms) that can learn from data, identify patterns, and make predictions or decisions with minimal human

Ask Gemini

+ Tools

Gemini can make mistakes, so double-check it

Type here to search

33°C Mostly cloudy

3:30 PM 21/10/2025